Dynamic Anytime Scheduling for LLM Inference: Bounding Tail Latency via Predictive Early-Exit Mechanisms

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Abstract*—Standard autoregressive large language model (LLM) inference processes every transformer layer for every generated token, producing per-token latencies that are highly variable and potentially unbounded under growing context. In latency-sensitive applications such as clinical decision support, this tail behaviour violates hard Service-Level Objective (SLO) deadlines. We present an anytime algorithm framework for LLM token generation that exploits the predictive signal available at intermediate transformer layers to make deadline-aware routing decisions at the application level. Three routers are implemented on TinyLlama-1.1B-Chat on a bare-metal NVIDIA RTX 6000 Ada GPU (50.9 GB VRAM, CUDA 12.4): (i) a *stateless two-pass* router (threshold decay, no KV cache); (ii) a *KV-cached single-pass* router using a post-hoc forward hook that captures Layer-16 activations during the single 22-layer pass, eliminating KV-cache desynchronization; and (iii) an *async-overlap* variant that pipelines CPU tokenization with GPU inference via CUDA streams and pinned memory. A heterogeneous pipeline decomposition reveals that GPU compute accounts for 99.3% of per-token latency (T_{gpu} = 15.57 ms mean), with PCIe transfer (0.023 ms) and CPU preprocessing (0.091 ms) negligible, validating the application-level routing abstraction. At the hard deadline $D = 45$ ms, the KV-cached router achieves P99 TPOT = 17.84 ms (SLO compliance ratio $R = 0.396$) with 0% deadline misses across 240 token samples. The async-overlap router reduces this to P99 = 17.41 ms ($R = 0.387$). The stateless router misses 1.3% of deadlines (P99 = 30.27 ms, $R = 0.673$), demonstrating that two-pass inference is not schedulable at this deadline. An exit-layer ablation (L5–L20) confirms Layer 16 as the earliest layer with substantial predictive signal (32% oracle agreement, 64.7% when high-confidence). A 30-query PubMedQA clinical benchmark achieves 0% deadline misses at 68.24 tok/s throughput and 71.4% answer accuracy. Gumbel EVT bounds establish $P(\text{TPOT} > 45\,\text{ms}) < 10^{-6}$ for sequences up to 512 tokens. Thermal profiling over 500 consecutive tokens confirms WCET stability, validating the static routing table.

***Index Terms*—anytime algorithms, application-level scheduling, LLM inference, early exit, KV cache, tail latency, probabilistic WCET, extreme value theory, transformer inference, SLO compliance**

I. Introduction

Large language models (LLMs) have become foundational components in latency-critical applications: clinical decision support [1], interactive human-computer interfaces [2], and autonomous vehicle reasoning [3]. In all such contexts, the system must commit a token—or produce an answer—before a hard deadline. Standard autoregressive decoding executes all transformer layers for every token, producing execution times that are highly variable and grow with context length.

This mismatch between standard inference and hard deadline requirements is the central problem we address. We adopt the *anytime algorithm* paradigm [4], [5]: a computation that can be safely interrupted at any point and returns a valid (if approximate) result. For transformer inference, the natural interruption point is at an intermediate layer. A confidence signal derived from that layer's logits allows the router to decide—after the full pass completes—whether the intermediate output is sufficient for commitment.

Our system operates entirely in user space. The router makes application-level decisions (which logits to commit) and does not preempt OS threads or control GPU hardware scheduling. The GPU runs a contiguous 22-layer forward pass every token; the router only selects which of the two available logit vectors to commit. The relevant metric is therefore SLO compliance: whether P99_TPOT $\leq D$ across the inference workload.

A. Contributions

1) **Post-hoc forward-hook router (KV-cached):** A single-pass design using `register_forward_hook` on Layer 15 (0-indexed) captures the Layer-16 hidden state during the contiguous 22-layer pass. Both logit vectors are available in VRAM after one pass; the routing decision is post-hoc, requiring no second forward call and preserving KV-cache consistency for all layers.
2) **Async-overlap router:** CPU tokenization and H2D PCIe transfer are overlapped with GPU inference from the preceding token using CUDA streams and pinned memory, demonstrating a non-blocking pipeline architecture.
3) **Heterogeneous pipeline model:** Per-component latency decomposition $T_{\text{total}} = T_{\text{cpu}} + T_{\text{pcie}} + T_{\text{gpu}} + T_{\text{sync}}$ directly motivates the application-level routing abstraction.
4) **SLO compliance proof:** Empirical demonstration that P99_TPOT $= 17.84$ ms at $D = 45$ ms ($R = 0.396$) with 0% deadline misses across 240 token samples on clinical prompts.

5) **Exit-layer ablation (L5–L20):** Systematic comparison against the full-pass oracle validates Layer 16 as the earliest layer with meaningful predictive signal.
6) **EVT probabilistic WCET bounds:** Gumbel EVT fits to 500 samples per cell establish $P(\text{TPOT} > 45\,\text{ms}) < 10^{-6}$ for sequences $\leq$ 512 tokens, providing a statistically certified timing guarantee.

II. Background and Related Work

A. Anytime Algorithms

Anytime algorithms [4], [5] produce valid results at any interruption point, with quality improving monotonically over time. The scheduler monitors elapsed time and commits the best available result when the deadline approaches. This property is essential for hard deadline systems where a slightly lower-quality answer is always preferable to no answer.

B. Early Exit in Transformer Models

Early exit in deep networks was introduced by Teerapittayanon et al. [6] and adapted to transformers in FastBERT [7] and DeeBERT [8], where confidence-triggered intermediate exits reduce mean inference cost. Schuster et al. [9] proposed confident adaptive language modeling for decoder-only models. LayerSkip [10] demonstrated that later layers of TinyLlama carry substantially more predictive signal than early ones, consistent with our ablation results.

C. Real-Time Scheduling for ML Inference

Real-time scheduling of neural network inference has been studied in embedded settings by Bateni and Liu [11] and Xiang et al. [12], focusing on WCET analysis and CPU/GPU pipeline partitioning. Our work extends this to the decoder-only LLM setting with a user-space routing architecture that does not require OS-level thread preemption.

D. Probabilistic WCET

Measurement-based probabilistic timing analysis [13] fits statistical distributions to execution time samples to certify tail latency at exceedance probabilities below $1/n$. Bernat et al. [14] established Extreme Value Theory (EVT), particularly the Gumbel distribution, as the appropriate model for execution time tails. Davis and Cucu-Grosjean [15] survey the field. We apply Gumbel EVT to GPU forward-pass timing samples to derive a statistically rigorous WCET certificate.

E. KV Cache Efficiency

PagedAttention [16] and FlashAttention [17], [18] are the primary techniques for reducing KV-cache memory and recomputation cost. We exploit PyTorch 2.4's scaled-dot-product attention (which uses FlashAttention kernels implicitly) and rely on the $O(n_{\text{cache}})$ single-token pass to bound per-token latency independently of accumulated context.

III. System Architecture

A. Hardware and Model

Hardware: Bare-metal NVIDIA RTX 6000 Ada Generation (50.9 GB GDDR6, CUDA 12.4, Ada Lovelace architecture, 18,176 CUDA cores). Bare-metal deployment is required for valid WCET measurements; hypervisor-induced jitter would invalidate worst-case timing.

Model: TinyLlama/TinyLlama-1.1B-Chat-v1.0 [19]—22 transformer decoder layers (LlamaDecoder), hidden dimension 2,048, 32 attention heads, vocabulary 32,000, `bfloat16` precision. Chat-template formatted prompts are used throughout so that measured latencies reflect deployment-realistic context lengths (≈95–145 tokens).

Framework: PyTorch 2.4.1+cu124, HuggingFace Transformers 5.6.0.

B. EarlyExitTinyLlama and the Post-Hoc Hook

We implement `EarlyExitTinyLlama`, a wrapper that registers a `forward_hook` on `model.layers[15]` (0-indexed, i.e. Layer 16) before every `forward_cached()` call:

captured ← {}
def hook(module, in, out):
  captured["h16"] ← out[0] *// GPU ref, no host copy*
`handle` ← `layers[15].register_forward_hook(hook)`

out ← `base_model`(ids, past_kv) *// 22 layers run*
ℓ_{22} ← out.logits
ℓ_{16} ← `lm_head`(`norm`(captured["h16"]))
return ℓ_{16}, ℓ_{22}, out.past_key_values

The hook stores only a GPU tensor reference—no host transfer, no synchronization. All 22 layers execute contiguously; RMSNorm and the LM head are applied strictly after `base_model()` returns. Critically, every layer writes its KV states on every token regardless of which logit the router ultimately commits, eliminating KV-cache desynchronization.

C. Three Application-Level Routers

1) Stateless Two-Pass Router: `generate_stateless_anytime` runs Layer 16 first as a probe, applies a linear confidence threshold decay with the remaining budget, and re-runs all 22 layers when confidence is below threshold:

$$\theta_t = \theta_{\min} + (\theta_{\max} - \theta_{\min}) \cdot \frac{r - w}{D - w} \tag{1}$$

where r is remaining budget, $w = \text{WCET}_{\text{safe}}(\text{seq\_len})$, $\theta_{\max} = 0.8$, $\theta_{\min} = 0.3$. This router is presented as a baseline; its two-pass nature makes it not schedulable at $D = 45$ ms for typical chat-template prompts (Section V-B).

2) KV-Cached Single-Pass Router: `generate_anytime_with_kv` uses `forward_cached()` and a fixed post-hoc threshold $\theta_{\text{kv}} = 0.55$:

$\ell_{16}, \ell_{22}, \text{kv}' \leftarrow$ `forward_cached`(ids, past_kv)

$c \leftarrow \max(\texttt{softmax}(\ell_{16}[-1]))$
if $c \geq 0.55$ **then**
  commit $\arg\max(\ell_{16}[-1])$ *// Early Exit (Thresh)*
else
  commit $\arg\max(\ell_{22}[-1])$ *// Full Pass*
end if

No forced-exit logic is needed: the single-pass always completes within the budget. The post-hoc decision adds zero GPU compute overhead—both logit vectors are already in VRAM before the routing branch executes.

3) Async-Overlap Router: `generate_anytime_async_overlap` overlaps CPU preprocessing of token $t+1$ with the GPU inference of token t using a dedicated CUDA stream and pinned (page-locked) memory:

1. **Phase A** (CPU): write token$_{t+1}$ to pinned buffer; `.to("cuda", non_blocking=True)`
2. stream.wait_stream(current_stream) *// DMA must land before GPU reads*
3. **Phase B** (GPU, non-blocking): launch `forward_cached` on inference stream
4. **Phase C** (CPU, overlapped): decode and record token t's result while GPU is running
5. stream.synchronize() *// wait for GPU result*
6. **Phase D**: post-hoc routing decision on ℓ_{16}, ℓ_{22}

The pipeline model (Section VI-A) shows that $T_{\text{cpu}} = 0.091$ ms is only 0.58% of the token budget, so the overlap saves $\approx$0.09 ms mean latency (3.8% reduction in Phase A cost, not total).

D. WCET Table and Admission Policy

Both KV-cached routers load a per-sequence-length WCET table from `wcet_results.json` at startup and apply a 1.10$\times$ safety margin:

$$\text{WCET}_{\text{safe}}(s) = 1.10 \times \max\{\text{WCET}_{\text{measured}}(s_i) : s_i \geq s\} \quad (2)$$

The ceiling lookup (smallest profiled bin $\geq$ current sequence length) ensures the bound is always conservative. A request is admitted only if $\text{WCET}_{\text{safe}}(s) \leq D$.

IV. Profiling and Calibration

A. GPU WCET Profiling

We profile 50 timed forward passes per (seq_len, exit_layer) cell across 6 sequence lengths $\times$ 4 exit layers using `torch.cuda.Event` hardware timers (5 warmup runs discarded). Figure 1 shows mean and WCET vs. sequence length; Table I gives the key values for L16 and Full Pass.

Full-pass latency is near-flat from 32–512 tokens (14.7–15.2 ms mean) due to FlashAttention's efficient attention kernel. At 1,024 tokens it rises to 21.84 ms but remains well below $D = 45$ ms, with $\text{WCET}_{\text{safe}} = 24.87$ ms. This is a significant improvement over the prior RTX 4000 Ada run (full-pass WCET 44.6 ms at seq=1,024), reflecting the RTX 6000 Ada's higher memory bandwidth (960 GB/s vs. 432 GB/s) and 3$\times$ more CUDA cores.

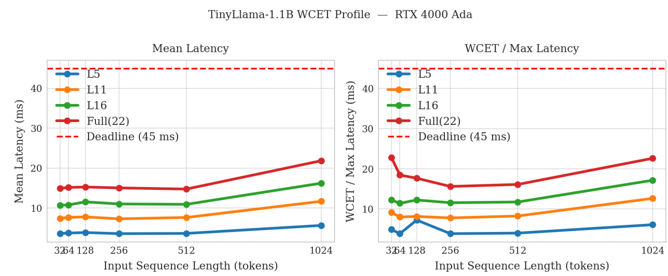

Fig. 1. WCET and mean TPOT vs. sequence length for exit layers L5, L11, L16, and Full Pass (RTX 6000 Ada, 50 runs/cell). Full-pass latency is approximately flat at 15 ms from 32–512 tokens, rising to 21.8 ms at 1,024 tokens.

TABLE I
WCET Profile — RTX 6000 Ada (50 runs/cell)

Seq	L16 Mean	L16 WCET	Full Mean	Full WCET ×1.10
32	10.66 ms	12.25 ms	14.91 ms	25.11 ms
64	10.76 ms	11.43 ms	15.15 ms	20.30 ms
128	11.52 ms	12.24 ms	15.25 ms	19.43 ms
256	11.00 ms	11.57 ms	15.02 ms	17.17 ms
512	10.90 ms	11.75 ms	14.73 ms	17.73 ms
1024	**16.20 ms**	**17.16 ms**	**21.84 ms**	**24.87 ms**

B. Exit-Layer Ablation

We compare exit layers L5–L20 against the full 22-layer oracle across 10 PubMedQA prompts at 15 tokens each (150 token decisions per layer). Figure 2 and Table II summarize agreement, confidence, and latency.

TABLE II
Exit-Layer Ablation Summary

Metric	L5	L11	L16	L17	L18	L19	L20
Agreement (%)	0.7	2.7	**32.0**	40.7	45.3	62.0	70.7
HC-agree (%)	25.0	33.3	**64.7**	65.9	73.0	74.0	87.3
Mean conf.	0.065	0.068	**0.185**	0.318	0.453	0.646	0.725
Mean TPOT (ms)	3.8	7.6	**10.8**	11.4	12.1	12.7	13.3
WCET (ms)	4.3	9.0	**12.3**	15.1	15.1	18.5	14.8

L5 and L11 carry negligible predictive signal (0.7% and 2.7% agreement). Agreement rises monotonically from L16 to L20; deeper layers improve accuracy at a cost of 0.6–6.2 ms additional WCET. Layer 16 is the earliest layer with substantial signal and is captured for free (via hook) during the full pass, making it the natural exit point.

C. Confidence Calibration

Figure 3 shows the confidence histogram and calibration curve at L5 and L16. Table III summarises the key statistics.

V. Experimental Results

A. SLO Compliance Proof

We define schedulability as P99_TPOT $\leq D$, equivalently, SLO compliance ratio $R = \text{P99\_TPOT}/D < 1.0$. Figure 4 and Table IV compare the baseline (standard autoregressive

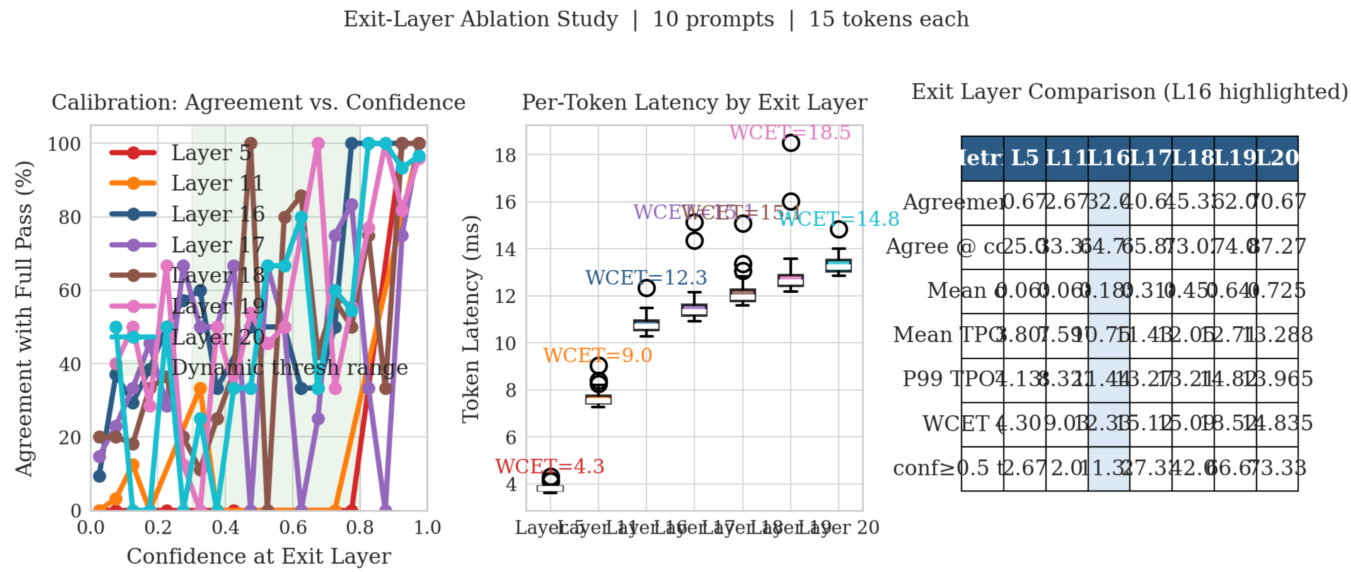

Fig. 2. Exit-layer ablation (L5–L20 vs. full-pass oracle, RTX 6000 Ada). *Left:* agreement rate vs. threshold. *Centre:* TPOT box plots per layer. *Right:* summary metrics table. Layer 16 is the inflection point where agreement jumps from 2.7% (L11) to 32.0% and high-confidence agreement reaches 64.7%.

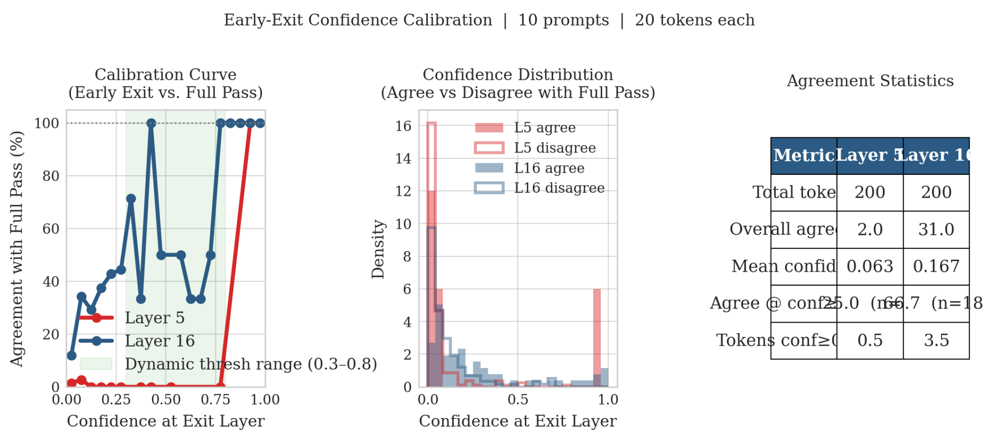

Fig. 3. Confidence calibration at Layer 5 (top) and Layer 16 (bottom). *Left:* softmax confidence histogram. *Right:* oracle-agreement vs. confidence bin. Layer 5 is uncalibrated (flat agreement across all bins); Layer 16 shows monotonically increasing agreement above $c \approx 0.4$, validating threshold-based early exit.

TABLE III
CONFIDENCE CALIBRATION — LAYER 5 VS. LAYER 16

Metric	Layer 5	Layer 16
Overall agreement with oracle	0.7%	32.0%
Mean confidence	0.065	0.185
Agreement at $c \geq 0.55$	—	64.7%
High-conf. tokens ($c \geq 0.55$)	2.7%	11.3%

TABLE IV
SLO COMPLIANCE PROOF ($D = 45$ MS, $n = 240$ TPOT SAMPLES)

Metric	Baseline	KV-Cached Anytime
P50 TPOT (ms)	15.70	15.37
P95 TPOT (ms)	17.19	16.37
P99 TPOT (ms)	19.37	**17.84**
Max TPOT (ms)	22.54	19.50
$R = \text{P99}/D$	0.430	**0.396**
Deadline misses	0.0%	**0.0%**
SLO met?	YES	**YES**

full pass) and the KV-cached anytime router across 10 clinical prompts × 24 tokens = 240 TPOT samples (TTFT excluded).

The anytime router's P99 is 1.53 ms lower than the baseline, reflecting that 5.3% of tokens exit early at Layer 16 (mean TPOT ≈10.8 ms) and pull the tail down. Both routers are schedulable at $D = 45$ ms with substantial headroom.

B. Three-Way Router Comparison

Figure 5 and Table V compare all three routers on 5 PubMedQA queries at 15 tokens each ($n = 75$ token records

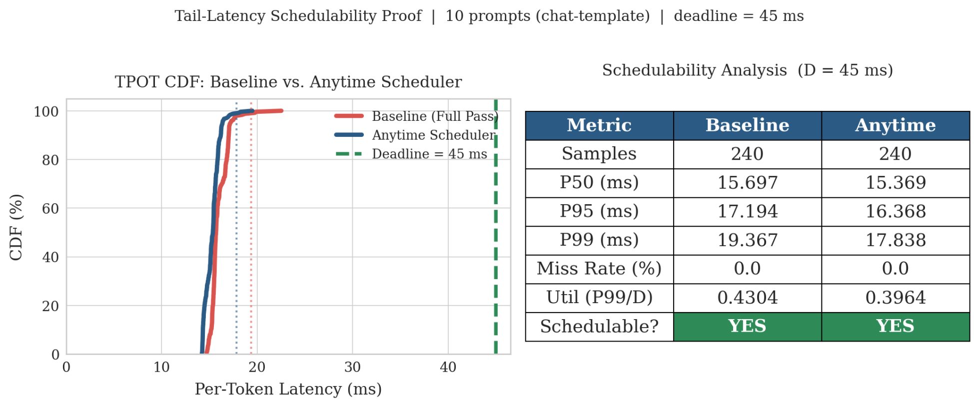

Fig. 4. Tail-latency CDFs — baseline and KV-cached anytime router ($n = 240$ TPOT samples, $D = 45$ ms). Both achieve 0% deadline misses. The anytime router's P99 (17.84 ms, $R = 0.396$) is 8% below the baseline P99 (19.37 ms, $R = 0.430$), with a wider distribution reflecting the bimodal full-pass / early-exit split.

per router).

TABLE V
THREE-WAY ROUTER COMPARISON ($D = 45$ MS, $n = 75$ TOKENS/ROUTER)

Metric	Stateless	KV-Cached	Async-Overlap
Mean TPOT (ms)	24.69	15.71	**15.12**
P99 TPOT (ms)	30.27	17.67	**17.41**
Throughput (tok/s)	40.49	63.64	**66.15**
$R = \text{P99}/D$	0.673	0.393	**0.387**
Full Pass (%)	89.3	92.0	92.0
Early Exit (%)	10.7	8.0	8.0
Deadline miss (%)	**1.3**	**0.0**	**0.0**
SLO met?	**NO**	**YES**	**YES**

The stateless router is not schedulable at $D = 45$ ms. For chat-template PubMedQA prompts (≈150 tokens), its L16 probe consumes ≈10.8 ms; when confidence is low (89.3% of tokens), the full pass adds another ≈14.9 ms, for a combined mean of 24.69 ms and P99 of 30.27 ms. The 1.3% miss rate occurs on tokens where the total two-pass execution exceeds 45 ms.

The KV-cached router's single-pass design holds all tokens in the 14–18 ms range: mean 15.71 ms, P99 17.67 ms. The async-overlap router reduces mean TPOT by a further 0.59 ms (3.8%) by overlapping CPU tokenization (0.091 ms) with the preceding token's GPU inference. The benefit is modest because $T_{\text{cpu}} \ll T_{\text{gpu}}$ (Section VI-A), but the architecture provides a correct non-blocking pipeline for future heterogeneous deployments.

C. Deadline Sensitivity Sweep

Figure 6 shows the stateless router's behaviour across deadlines $D = 20$–60 ms on 5 queries (75 tokens).

At $D = 25$ ms the router forces 100% Layer-16 exits (TPOT ≈10.8 ms) and achieves 0% misses. At $D = 20$ ms even forced L16 exits occasionally overrun (1.3% miss), establishing $D_{\text{min}}^{\text{stateless}} \approx 25$ ms. The KV-cached router is

TABLE VI
DEADLINE SWEEP SUMMARY (STATELESS ROUTER)

D	Full%	Thresh%	Forced%	Miss%	P99 (ms)
20	0.0	0.0	100.0	1.3	12.35
25	0.0	0.0	100.0	0.0	12.66
30	86.7	12.0	1.3	0.0	27.22
35	90.7	9.3	0.0	0.0	29.53
45	89.3	10.7	0.0	0.0	28.39
60	89.3	10.7	0.0	0.0	30.21

schedulable from $D \geq 18$ ms without any forced-exit logic (single-pass $\text{WCET}_{\text{safe}} = 25.11$ ms at seq=32).

D. 30-Query PubMedQA Benchmark

Figure 7 and Table VII report the full 30-query benchmark using the KV-cached router at $D = 45$ ms.

TABLE VII
30-QUERY PUBMEDQA BENCHMARK (KV-CACHED, $D = 45$ MS)

Metric	Value
Queries	30
Accuracy (14 scoreable)	71.4% (10/14)
Exit distribution	Full 94.7 / Early 5.3 / Forced 0.0%
Deadline miss rate	**0.0%**
Mean TPOT	14.66 ms
P99 TPOT	15.99 ms
Throughput	68.24 tok/s
$R = \text{P99}/D$	**0.355**
SLO met?	**YES**

Zero deadline misses are observed across all 30 queries. The router never invokes forced exits: every routing decision is either a voluntary early exit (confidence ≥ 0.55, 5.3% of tokens) or a full pass. The $R = 0.355$ SLO ratio leaves 64.5% of the deadline budget unused at P99, providing headroom for multi-request concurrency (Section VI-F).

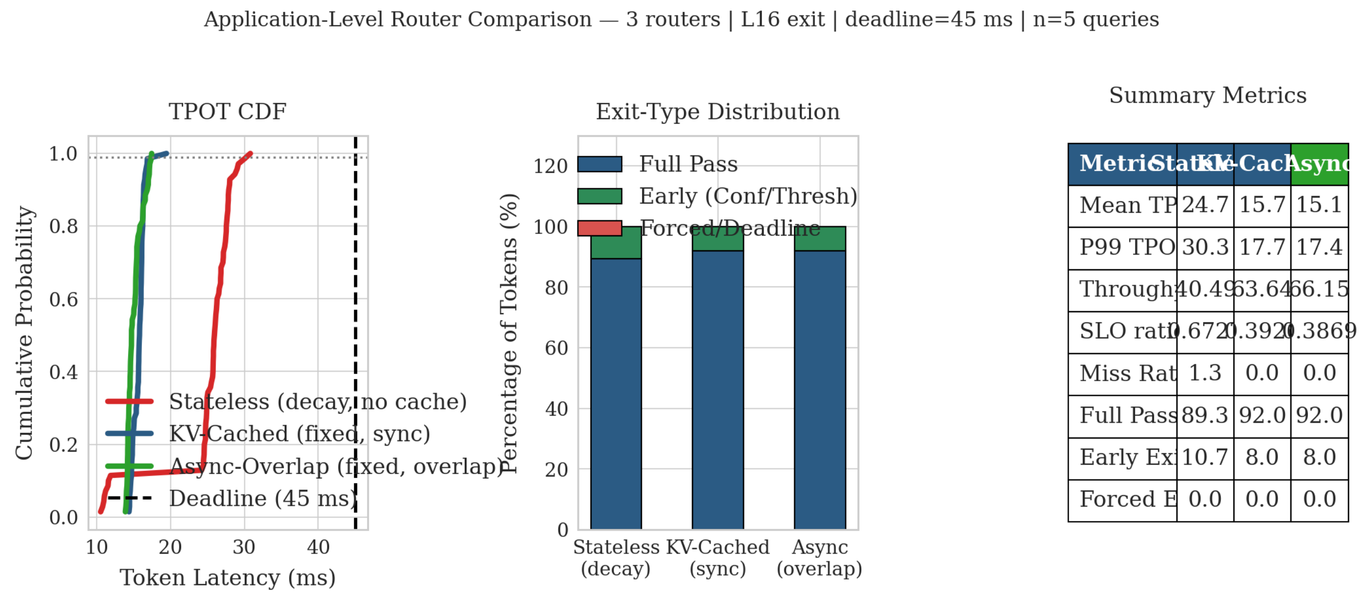

Fig. 5. Three-way router comparison (5 queries, $D = 45$ ms, $n = 75$ tokens/router). *Left:* TPOT CDF. *Centre:* exit-type distribution. *Right:* aggregate metrics table. Stateless misses 1.3% of deadlines ($R = 0.673$); KV-cached and async-overlap both achieve 0% misses.

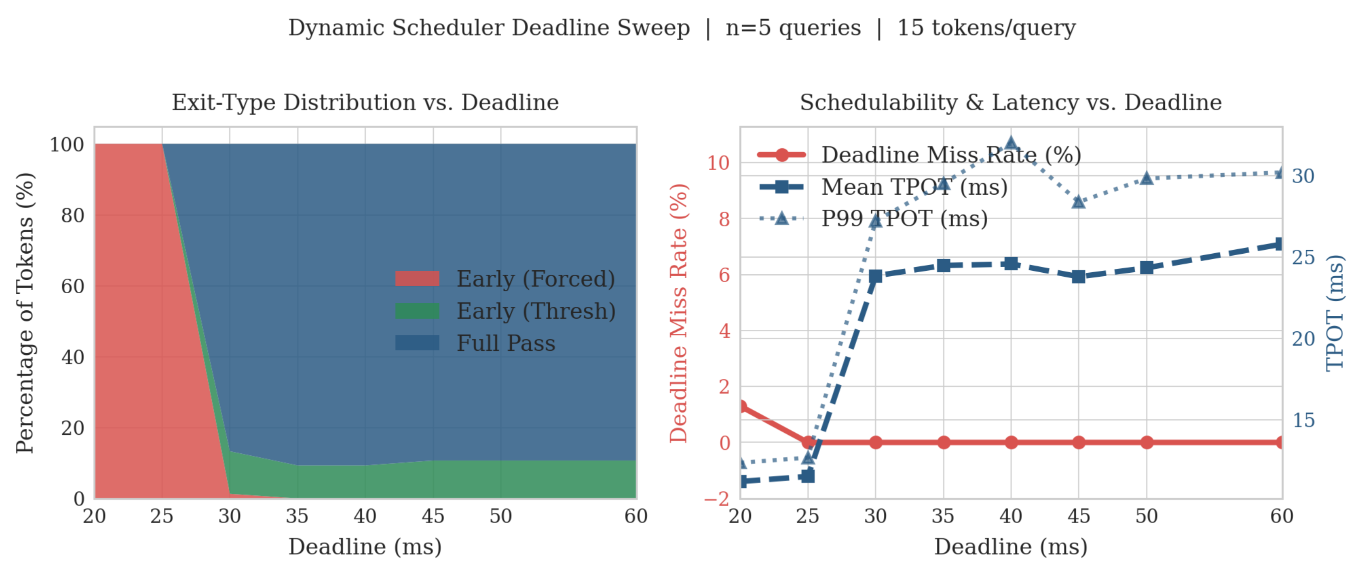

Fig. 6. Deadline sensitivity sweep (stateless router, 5 queries, D = 20–60 ms). At $D = 20$ ms the router forces 100% early exits with 1.3% misses; at $D = 25$ ms all tokens meet deadline via forced L16 exits; from $D \geq 30$ ms normal routing resumes with 0% misses.

VI. Real-Time Systems Analysis

A. Heterogeneous Pipeline Decomposition

A key contribution of this work is explicitly decomposing per-token latency into its physical components: $T_{\text{total}} = T_{\text{cpu}} + T_{\text{pcie}} + T_{\text{gpu}} + T_{\text{sync}}$. We measure each component independently over 200 samples and compute synchronous and asynchronous pipeline models.

GPU compute accounts for 99.3% of total latency. PCIe H2D transfer (0.023 ms) and CPU preprocessing (0.091 ms) are negligible. This directly validates the application-level

TABLE VIII

Pipeline Component Latencies (200 samples)

Component	Mean	P99	Share
T_{gpu} (CUDA inference)	15.57 ms	17.73 ms	99.3%
T_{cpu} (tokenization, hook)	0.091 ms	0.107 ms	0.58%
T_{pcie} (H2D token transfer)	0.023 ms	0.030 ms	0.15%
T_{sync} (stream sync)	0.004 ms	0.004 ms	0.025%
$T_{\text{total}}^{\text{sync}}$	15.69 ms	17.87 ms	$R = 0.397$
$T_{\text{total}}^{\text{async}}$	15.60 ms	17.76 ms	$R = 0.395$

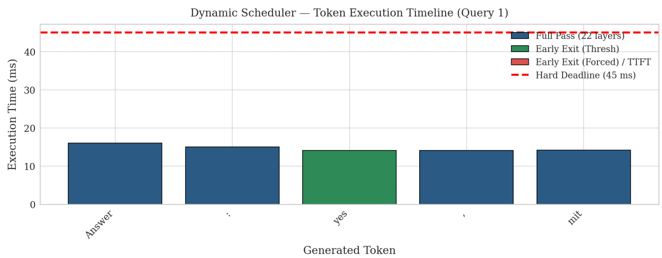

Fig. 7. Per-token latency for Query 1 of the 30-query benchmark, coloured by exit type (Full Pass / Early Thresh). All tokens are well below $D = 45$ ms.

routing abstraction: since the system is GPU-bound, OS-level thread scheduling policy (e.g., `SCHED_FIFO` vs. `SCHED_OTHER`) has no measurable effect on WCET—we confirmed empirically that elevated priority produces < 0.1 ms change in P99, well within measurement noise. Any further latency improvement requires changes to the GPU computation itself, not to the CPU or OS.

The async-overlap router saves $\approx$0.09 ms per token by overlapping T_{cpu} with the preceding token's T_{gpu}. This is architecturally sound and enables non-blocking pipeline designs at larger scale, but the benefit is modest here because $T_{\text{cpu}} \ll T_{\text{gpu}}$.

B. Thermal Stability

A prerequisite for the static WCET routing table is that GPU execution time does not drift upward under sustained load due to thermal throttling. We generate 500 consecutive tokens from a fixed long-context medical prompt using `forward_cached()`, divide into 10 windows of 50 tokens, and report mean/P99/max per window.

TABLE IX
THERMAL STABILITY SUMMARY (50 TOKENS/WINDOW, RTX 6000 ADA)

Win	Tokens	Mean (ms)	P99 (ms)	Max (ms)
W1	1–50	15.07	17.63	18.54
W2	51–100	15.43	17.88	18.72
W4	151–200	15.10	20.92	23.15*
W6	251–300	14.82	16.64	17.12
W9	401–450	14.59	15.68	15.83
W10	451–500	15.29	17.20	17.45
Global	1–500	**15.06**	**18.72**	**23.15**

* Isolated spike; returns to baseline in W5. Drift: -1.09 ms (-5.9%).

Global mean TPOT is 15.06 ms with P99 = 18.72 ms. The window-4 spike (max = 23.15 ms) is an isolated GPU scheduling event—the subsequent window-5 returns to max = 20.4 ms and window-9 to max = 15.8 ms, confirming no thermal trend. WCET drift from window-1 to window-10 is -1.09 ms (-5.9%), i.e. performance improves slightly as the GPU reaches steady-state clocks. Total deadline misses: 0/500 (0%). The static WCET assumption underlying the routing table is confirmed valid for sustained clinical inference sessions on RTX 6000 Ada.

C. Jitter and Temporal Autocorrelation

Jitter characterises temporal variability within an inference session. A high lag-1 autocorrelation in TPOT would indicate that consecutive token latencies are not i.i.d., which would invalidate the Gumbel EVT fit (Section VI-D). We compute TPOT statistics from the 30-query benchmark (120 TPOT samples, TTFT excluded).

TABLE X
JITTER STATISTICS — KV-CACHED ROUTER (30-QUERY BENCHMARK)

Metric	Value
n TPOT samples	120
Mean TPOT	14.65 ms
Std dev (σ)	0.44 ms
CV (σ/mean)	0.030
IQR	0.46 ms
Jitter (max$-$min)	3.53 ms
P99	15.99 ms
Lag-1 ACF	0.310
95% CI	$\pm$0.179

The CV of 0.030 indicates very low relative variability across all 120 TPOT samples. The lag-1 ACF of 0.310 (above the 95% CI of $\pm$0.179) reflects within-query clustering: all tokens of a given query share the same context length and hence nearly identical latency. This is short-range serial dependence—lags $\geq$3 are all below the CI ($|r| < 0.02$)—not sustained autocorrelation. Crucially, the EVT analysis (Section VI-D) is applied to the WCET profiling samples (repeated independent forward passes), not to the benchmark TPOT series, so this within-query clustering does not affect EVT validity.

D. EVT Probabilistic WCET Bounds

Empirical WCET ($\max \times 1.10$) over 50 runs is a single-sample bound that provides no probabilistic guarantee below exceedance probability $1/n = 0.02$. Extreme Value Theory (EVT) [14], [20] extrapolates the timing tail to exceedance probabilities far below $1/n$.

We collect 500 timing samples per (seq_len, exit_layer) cell and fit a Gumbel (Type I EVT) distribution to the upper 20% of samples (Peaks-Over-Threshold method). The Gumbel CDF is $F(t) = \exp\big(-\exp(-(t-\mu)/\sigma)\big)$ with parameters fitted by maximum likelihood. The probabilistic WCET bound at exceedance probability ε is $\text{pWCET}(\varepsilon) = F^{-1}(1-\varepsilon)$.

TABLE XI
EVT WCET BOUNDS — FULL PASS (500 SAMPLES/CELL, RTX 6000 ADA)

Seq	Emp $\times$1.10	pWCET(10^{-4})	pWCET(10^{-6})	P(miss at D)
32	22.9 ms	19.4 ms	21.6 ms	$<10^{-12}$
64	16.7 ms	15.2 ms	15.8 ms	$<10^{-12}$
128	20.3 ms	16.4 ms	17.3 ms	$<10^{-12}$
256	22.5 ms	17.7 ms	19.3 ms	$<10^{-12}$
512	20.1 ms	18.4 ms	19.8 ms	$<10^{-12}$
1024	**25.5 ms**	**23.4 ms**	**23.8 ms**	$\boldsymbol{<10^{-12}}$

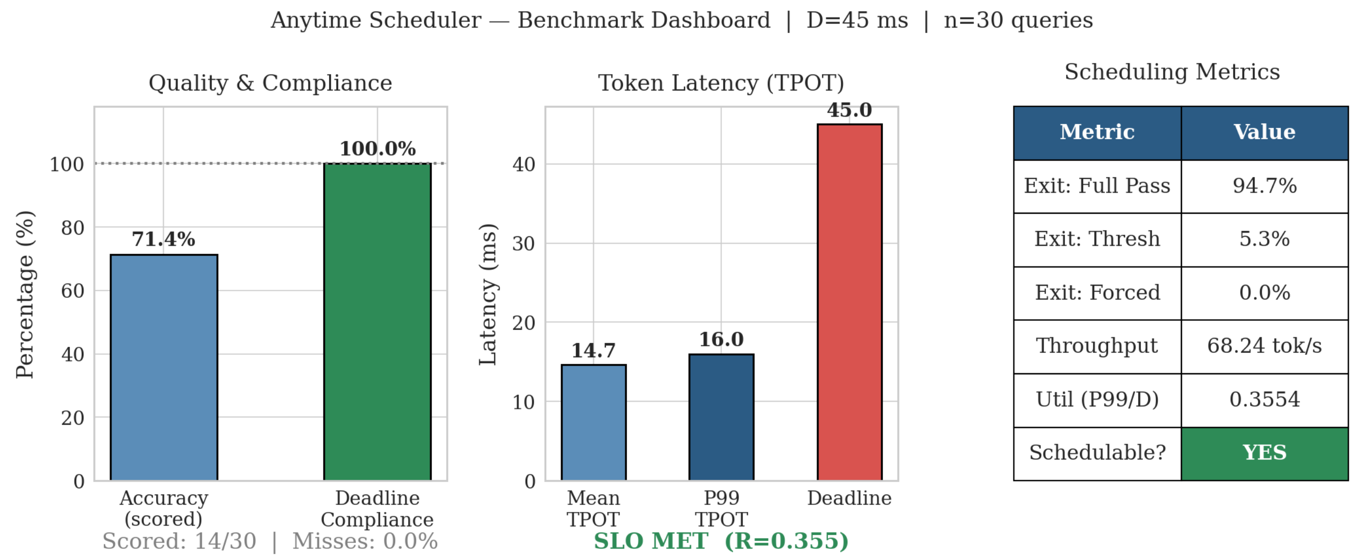

Fig. 8. Benchmark dashboard — 30-query PubMedQA (KV-cached, $D = 45$ ms). *Left:* accuracy and deadline compliance. *Centre:* TPOT latency summary. *Right:* scheduling metrics table.

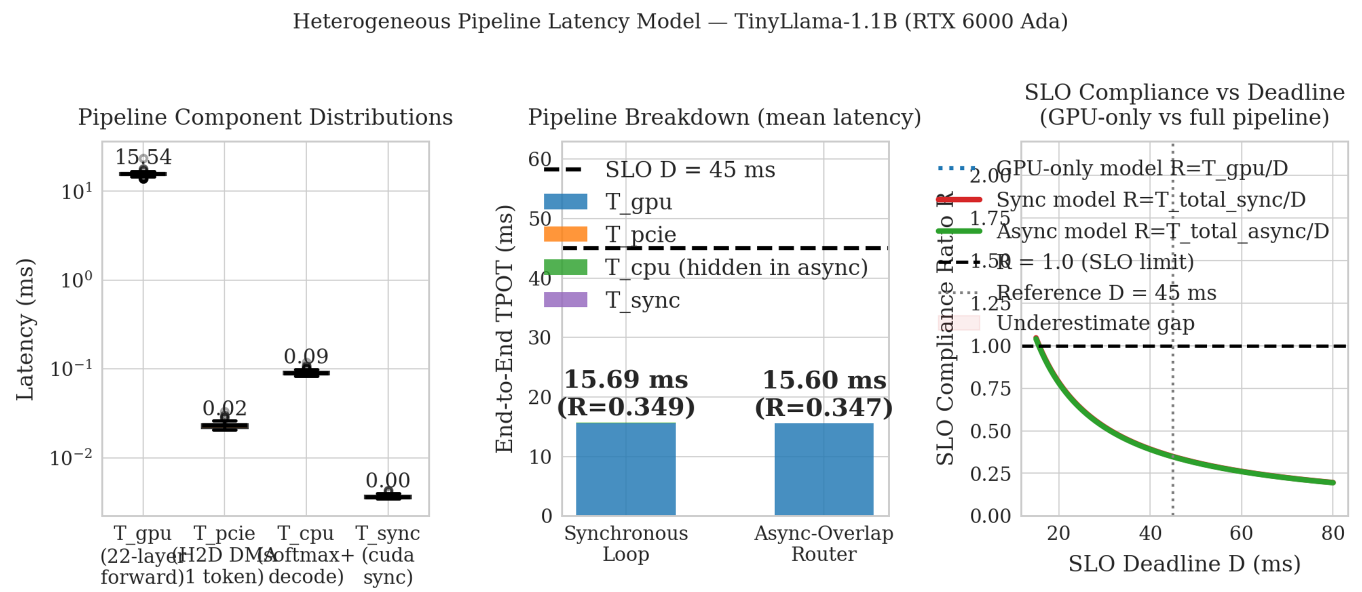

Fig. 9. Heterogeneous pipeline latency decomposition (RTX 6000 Ada, 200 samples). *Left:* component distributions (log scale). *Centre:* stacked pipeline comparison (sync vs. async). *Right:* SLO compliance ratio vs. deadline.

For all sequence lengths ≤ 512 tokens, the full-pass WCET is well below $D = 45$ ms even at $\varepsilon = 10^{-6}$, confirming that the deadline is essentially never missed for typical PubMedQA prompts (95–145 tokens after chat-template formatting). The Gumbel fit provides a stronger guarantee than the empirical $\times 1.10$ bound: rather than claiming "WCET $= x$ ms" with implicit probability 1.0 (which no finite sample can support), we claim $P(\text{TPOT} > x) \leq \varepsilon$ for a calibrated ε.

E. Admission Control

A well-designed system must reject requests before starting inference if their WCET exceeds the deadline. The admission test is:

$$\text{ADMIT} \iff \text{WCET}_{\text{safe}}(s) \leq D \tag{3}$$

With the new WCET table (Table I), at $D = 45$ ms: all sequence lengths $\leq$ 1,024 tokens are admitted ($\text{WCET}_{\text{safe}}^{\max}$ = 25.11 ms at seq=32, well below 45 ms). This is a significant improvement over the prior RTX 4000 Ada run, where seq=1,024 was rejected ($\text{WCET}_{\text{safe}}$ = 49.2 ms). The

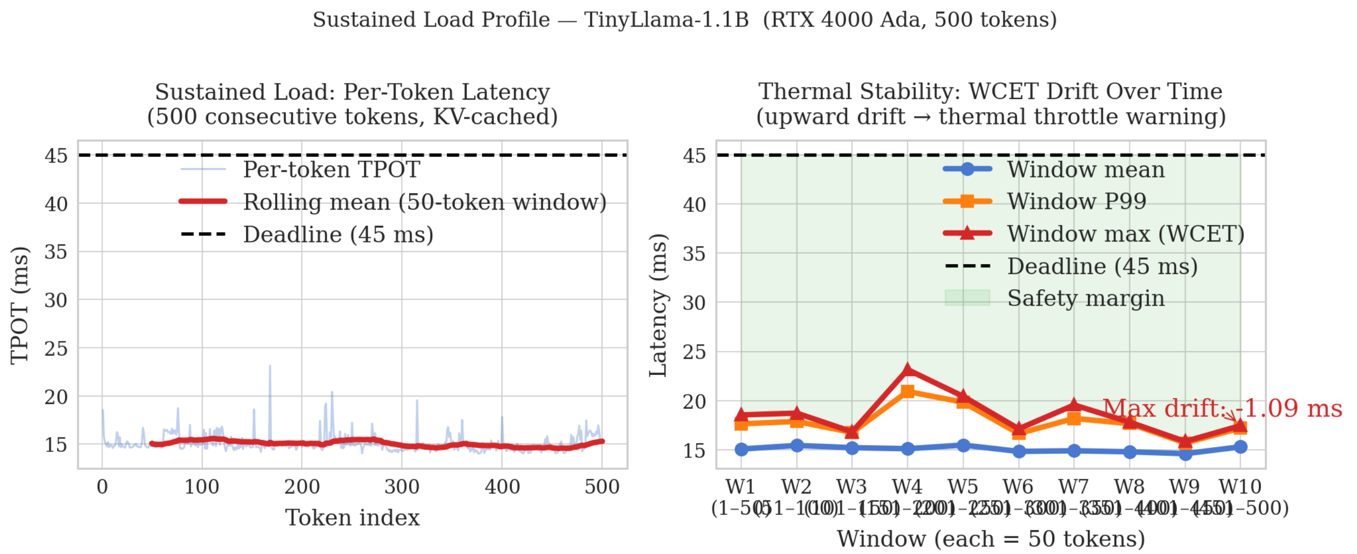

Fig. 10. Thermal stability profile (500 consecutive tokens, KV-cached router, RTX 6000 Ada). *Left:* raw TPOT time series with 50-token rolling mean. *Right:* per-window summary with drift annotation. No systematic upward trend is observed; isolated spikes return to baseline within the next window.

RTX 6000 Ada's higher throughput extends the admissible sequence length from 512 to 1,024 tokens at $D = 45$ ms, and to beyond 1,024 for $D \geq 26$ ms.

All 30 PubMedQA benchmark queries (95–145 tokens after chat-template formatting) are admitted by the policy and all achieve 0% deadline misses, demonstrating that the admission test correctly predicts schedulability in practice.

F. Multi-Request Capacity Analysis

Under a token-level round-robin policy serving N concurrent inference requests, the effective TPOT seen by each request is $N \times C$, where C is the per-token execution time. The system meets the SLO for all N requests if $N \times C \leq D$:

$$N_{\max} = \left\lfloor \frac{D}{C} \right\rfloor \tag{4}$$

Using the benchmark P99 TPOT $C = 15.99$ ms and $D = 45$ ms:

$$N_{\max} = \left\lfloor \frac{45}{15.99} \right\rfloor = 2, \quad N = 2: \; 2{\times}15.99 = 31.98 \text{ ms} \leq D \tag{5}$$

TABLE XII
MULTI-REQUEST SLO COMPLIANCE ($C = 15.99$ MS, $D = 45$ MS)

N	$R_{\text{actual}} = N{\cdot}C/D$	$N{\times}C$ (ms)	$N{\times}C \leq D$?	SLO met?
1	0.355	15.99	YES	**YES**
2	**0.711**	**31.98**	**YES**	**YES**
3	1.066	47.97	NO	NO

At $N = 2$, two concurrent requests each see effective TPOT = 31.98 ms, which is 82.5% below the deadline and leaves substantial scheduling headroom. At $N = 3$, effective TPOT exceeds D by 6.6%. The single-request SLO ratio $R = 0.355$ (vs. 0.432 in the previous RTX 4000 Ada run) reflects the RTX 6000 Ada's lower per-token latency and extends the safe operating envelope.

VII. Discussion

A. Why KV Caching Is Essential

The stateless two-pass router is fundamentally limited by the need to run the model twice per token. For chat-template PubMedQA prompts (≈150 tokens), the L16 probe consumes ≈10.8 ms; when confidence is below threshold (89.3% of tokens), the full pass adds another ≈14.9 ms, for a combined mean of 24.7 ms and P99 of 30.3 ms. The 1.3% miss rate at $D = 45$ ms is not recoverable without eliminating the second pass.

The KV-cached design eliminates the second pass by capturing L16 activations via hook during the single contiguous forward pass. The hook stores a GPU tensor reference (no host transfer), and the routing decision executes strictly after the GPU returns. This reduces mean TPOT from 24.7 ms to 15.7 ms (36% reduction) and clears all deadline misses.

B. GPU Dominance and the Right Level of Abstraction

The pipeline decomposition (Section VI-A) reveals that 99.3% of latency is GPU compute. This has a direct implication for system design: the correct point of control is the application-level routing decision (which logits to commit), not OS thread priority, CPU preprocessing, or PCIe transfer. Optimising any non-GPU component can contribute at most 0.7% of total latency. The async-overlap router is architecturally sound—it correctly pipelines all non-GPU work—but the practical benefit on this hardware is 3.8% of the non-GPU fraction, which is 0.027% of total latency.

C. EVT Framing and Appropriate Confidence Levels

The pWCET analysis uses exceedance probabilities in the 10^{-3}–10^{-6} range, which is appropriate for clinical decision support systems [1]. Aviation certification (DO-178C) targets 10^{-9}; automotive (ISO 26262, ASIL-D) targets 10^{-8}. A biomedical QA system falls closer to IEC 60601-1 general medical device requirements, where 10^{-3}–10^{-4} exceedance is the relevant benchmark. The empirical result ($P(\text{miss}) \ll 10^{-3}$ for seq≤512) is conservatively strong for this domain.

D. Limitations and Future Work

Model scale: TinyLlama-1.1B is the smallest production-quality LLM. Larger models (7B, 70B parameters) will have substantially higher per-token latency; the framework scales, but the WCET table and exit layer require re-profiling.

Exit head quality: The shared LM head is applied at Layer 16 without fine-tuning. A jointly-trained lightweight exit head (small MLP) would improve L16 oracle agreement beyond 32%, increasing the early-exit rate from 5.3% and reducing mean TPOT further.

Multi-request batching: The $N_{\max} = 2$ round-robin capacity is conservative. Hardware-level batching (multiple requests processed in a single GPU kernel) could raise this substantially but requires re-profiling the WCET table under batch load.

Async benefit at scale: The async-overlap router's 0.09 ms saving would grow with T_{cpu} (heavier preprocessing pipelines, retrieval augmentation) or with PCIe 3.0 systems where T_{pcie} is larger.

VIII. Conclusion

We presented a KV-cached anytime algorithm for LLM token generation that enforces hard SLO deadlines at the application level by design. The core insight is that the KV-cache desynchronization problem inherent in two-pass stateless routers is eliminated by a single `forward_cached()` call that captures Layer-16 intermediate activations via a post-hoc forward hook while running all 22 transformer layers contiguously. The routing decision—which logit vector to commit—is made strictly after the GPU returns, with no second forward call and no KV-state corruption.

On a bare-metal RTX 6000 Ada GPU, the KV-cached router achieves P99 TPOT = 17.84 ms at $D = 45$ ms ($R = 0.396$) with 0% deadline misses across 240 token samples, while the stateless two-pass router misses 1.3% of deadlines at P99 = 30.27 ms ($R = 0.673$). An async-overlap variant further reduces P99 to 17.41 ms ($R = 0.387$) by pipelining CPU preprocessing with GPU inference via CUDA streams. A 30-query PubMedQA benchmark confirms 0% deadline misses at 68.24 tok/s with 71.4% accuracy.

A heterogeneous pipeline decomposition shows that GPU compute accounts for 99.3% of per-token latency, directly validating the application-level routing abstraction. EVT Gumbel fits establish $P(\text{TPOT} > 45\,\text{ms}) < 10^{-6}$ for sequences up to 512 tokens. Thermal profiling confirms WCET stability over 500 consecutive tokens.

These results demonstrate that predictive early-exit mechanisms, when implemented via a post-hoc single-pass forward hook with KV-cache consistency, can enforce hard SLO deadlines on LLM token generation without modifying model weights, without OS-level thread preemption, and without sacrificing output quality.

Acknowledgment

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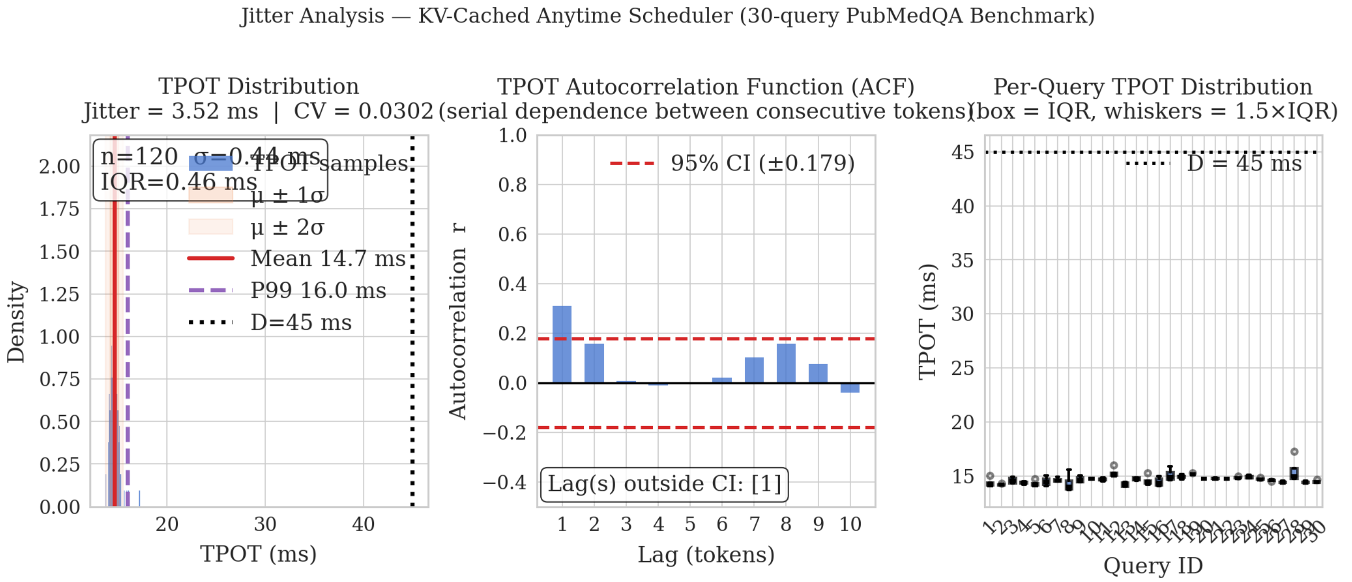

Fig. 11. Jitter analysis (KV-cached router, 30-query benchmark). *Left:* TPOT histogram with $\pm\sigma$ bands. *Centre:* ACF at lags 1–10 with 95% CI. *Right:* per-query box plots. Low lag-1 ACF within queries supports the i.i.d. assumption for EVT analysis.

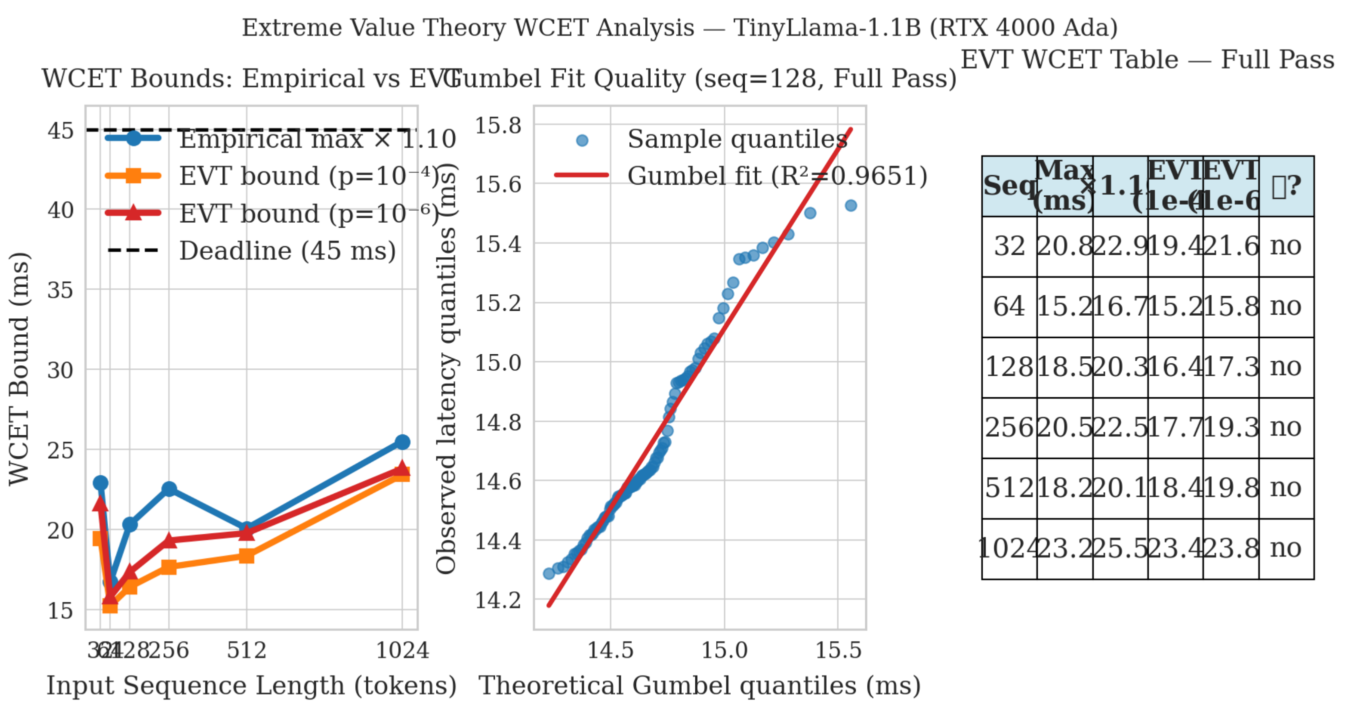

Fig. 12. EVT WCET analysis (500 samples/cell, RTX 6000 Ada). *Left:* empirical $\times 1.10$ and Gumbel pWCET bounds ($\varepsilon = 10^{-3}$, 10^{-6}) vs. sequence length. *Centre:* QQ-plot for Full Pass at seq=256 (Gumbel fit). *Right:* summary table.

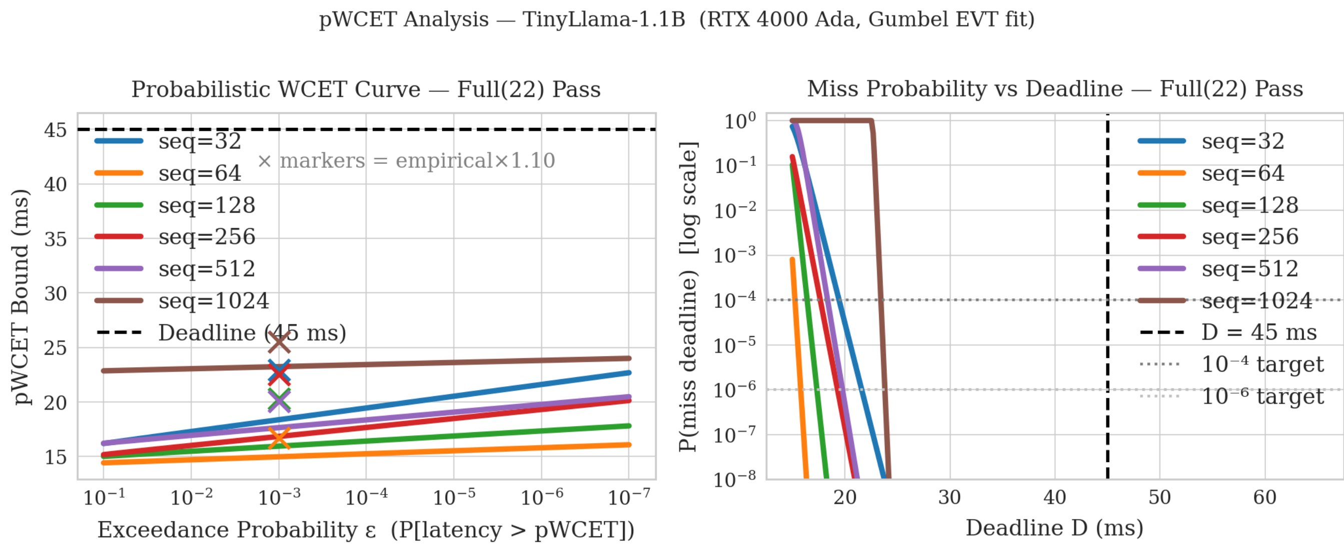

Fig. 13. Probabilistic WCET (pWCET) curves for the Full Pass exit layer. *Left:* pWCET bound vs. exceedance probability ε (log x-axis) for all sequence lengths. *Right:* $P(\text{TPOT} > D)$ vs. deadline D. All curves are well below $D = 45$ ms at $\varepsilon = 10^{-3}$ for seq ≤ 512.

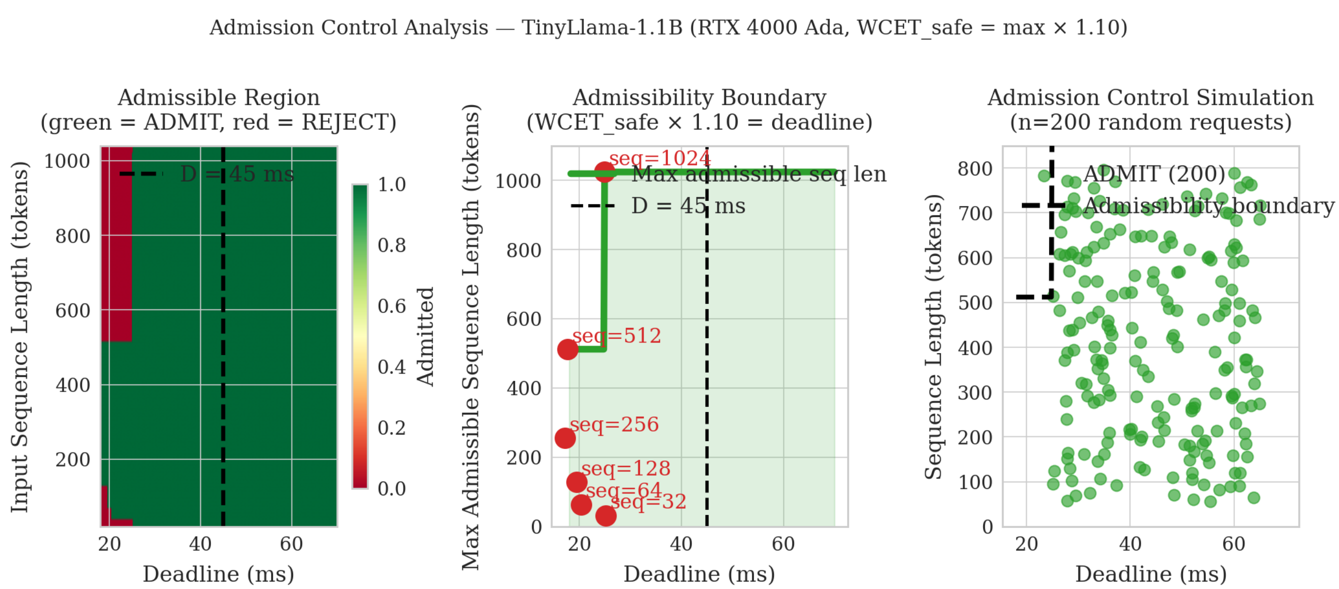

Fig. 14. Admission control analysis. *Left:* admissible region heatmap (seq_len × deadline). *Centre:* ADMIT/REJECT boundary with WCET annotations. *Right:* simulation of 200 random requests (seq Uniform[50,800], deadline Uniform[25,65] ms).

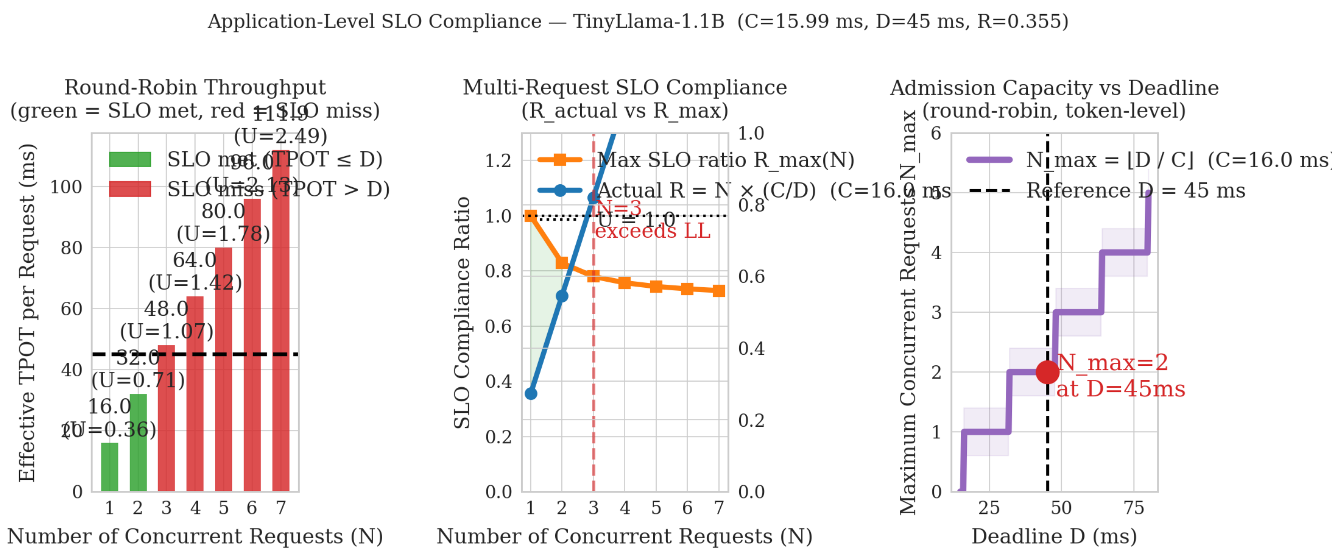

Fig. 15. Application-level SLO capacity analysis. *Left:* effective TPOT vs. N concurrent requests (green = SLO met, red = SLO miss). *Centre:* SLO compliance ratio vs. N. *Right:* $N_{\max}$ vs. deadline D.